

SANNIKOVA, Russian Federation

WMO: 215350

Lat: 74.667N

Lon: 138.900E

Elev: 48

StdP: 14.67

Time Zone: 10.00 (E10)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.4	-37.9	-44.7	0.4	-38.7	-42.3	0.5	-36.6	42.4	-5.1	36.6	-11.1	6.9	90	1.179

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	48.0	45.1	44.7	42.8	42.6	40.9	45.6	47.4	43.0	44.4	41.1	42.5	12.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
44.0	42.5	46.9	41.7	38.9	44.1	39.6	35.9	41.9	17.9	47.6	16.7	44.6	15.7	42.6	55.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.6	29.0	26.0	DB	-43.9	53.8	2.4	5.8	-45.6	58.0	-47.0	61.4	-48.4	64.6	-50.1	68.8	(4)
(5)			WB	-44.0	49.3	2.4	2.2	-45.7	50.9	-47.0	52.2	-48.4	53.4	-50.1	55.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-21.9	-22.8	-17.0	-1.4	18.0	33.0	36.6	36.7	31.8	16.6	-5.1	-17.0	(6)	
	DBStd	24.45	8.86	10.19	10.31	10.08	7.49	4.56	3.47	4.18	5.20	10.57	11.05	9.37	(7)	
	HDD50	15529	2228	2039	2078	1542	993	510	417	413	545	1035	1653	2076	(8)	
	HDD65	21004	2693	2459	2543	1992	1458	960	882	878	995	1500	2103	2541	(9)	
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	11.8	10.6	10.8	11.3	12.7	13.6	12.4	12.1	12.1	12.4	12.2	11.0	10.6	(14)	
	PrecAvg	10.9	0.5	0.5	0.5	0.4	0.6	0.9	1.5	1.7	1.5	1.3	0.9	0.6	(15)	
(16) Precipitation	PrecMax	15.0	1.2	1.6	1.9	1.6	1.7	2.4	2.5	3.5	2.7	3.0	1.7	1.9	(16)	
	PrecMin	6.5	0.1	0.0	0.0	0.0	0.1	0.2	0.6	0.6	0.4	0.4	0.2	0.1	(17)	
	PrecStd	2.0	0.3	0.4	0.4	0.4	0.4	0.5	0.5	0.7	0.6	0.6	0.4	0.4	(18)	
	PrecStd	2.0	0.3	0.4	0.4	0.4	0.4	0.5	0.5	0.7	0.6	0.6	0.4	0.4	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	8.7	19.8	24.1	35.0	53.2	54.5	52.1	43.3	34.5	25.4	8.9	(19)	
		MCWB	5.6	8.1	18.7	23.0	33.1	48.6	50.3	49.1	41.1	33.2	24.2	8.0	(20)	
	2%	DB	0.2	2.8	15.1	20.0	32.8	46.3	47.5	47.2	41.1	33.0	20.9	3.9	(21)	
		MCWB	-0.3	2.4	14.4	18.9	31.6	43.0	44.6	45.1	39.5	31.9	19.9	3.5	(22)	
	5%	DB	-4.0	-3.2	4.7	17.0	31.5	41.7	44.0	44.7	39.7	31.7	16.0	-0.7	(23)	
		MCWB	-4.3	-3.7	4.2	15.9	30.5	39.4	41.8	43.1	38.3	30.7	15.2	-1.3	(24)	
	10%	DB	-8.0	-8.1	-2.7	13.8	29.6	38.4	41.7	43.0	38.4	29.8	11.2	-3.6	(25)	
		MCWB	-8.4	-8.5	-3.2	12.9	28.5	36.6	39.8	41.6	37.2	28.6	10.6	-4.1	(26)	
	0.4%	WB	5.9	8.1	18.7	23.3	33.3	48.5	49.5	49.3	41.6	33.7	24.2	8.1	(27)	
		MCDB	6.3	8.5	19.4	23.9	34.2	52.8	52.9	51.8	43.2	34.2	25.1	8.8	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-0.1	2.6	14.3	19.0	32.0	43.1	44.7	45.5	39.7	32.1	20.1	3.4	(29)	
		MCDB	0.3	2.9	15.0	19.9	32.6	46.2	46.8	46.8	40.8	32.8	21.1	3.9	(30)	
	5%	WB	-4.5	-3.6	4.2	16.0	30.6	39.6	42.0	43.4	38.5	30.7	15.2	-1.1	(31)	
		MCDB	-4.3	-3.3	4.6	16.9	31.5	41.6	43.7	44.3	39.5	31.5	16.0	-0.7	(32)	
	10%	WB	-8.4	-8.4	-3.1	12.9	28.5	36.7	40.0	41.8	37.3	28.8	10.7	-4.0	(33)	
		MCDB	-8.0	-8.1	-2.8	13.8	29.6	38.2	41.5	42.9	38.2	29.8	11.2	-3.6	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.9	10.9	13.2	7.8	5.4	6.3	5.3	5.1	7.8	9.4	9.2	(35)	
		MCDBR	13.7	15.0	17.6	15.4	7.3	10.9	11.8	9.2	5.1	5.0	11.4	13.3	(36)	
		MCWBR	13.6	14.9	17.1	14.9	7.3	9.2	9.7	7.7	4.6	5.1	11.3	13.2	(37)	
	5% WB	MCDBR	13.7	14.9	17.5	15.3	7.4	10.8	11.2	8.4	4.9	4.8	11.8	13.2	(38)	
		MCWBR	13.6	14.8	17.2	14.9	7.4	9.2	9.6	7.2	4.6	5.0	11.7	13.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.169	0.268	0.301	0.331	0.322	0.354	0.344	0.288	N/A	N/A	N/A	(40)	
	taud		N/A	1.878	1.987	2.012	2.013	2.206	2.281	2.376	2.361	N/A	N/A	N/A	(41)	
	Ebn at Noon		0	95	194	243	255	266	250	234	203	N/A	0	0	(42)	
	Edh at Noon		0	6	22	34	40	35	31	24	16	N/A	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0	33	429	1245	1952	2091	1638	864	370	69	0	0	(44)	
	RadStd		0	5	23	54	70	109	89	103	31	7	0	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page